

Job Aggregation Platform — Final Project Report

Internal Engineering Report · Data Platform

Prepared for	Engineering Manager
Prepared by	Reuben Jacob
Report date	July 15, 2026
Data snapshot as of	July 10, 2026
Companion document	<code>Final_Project_Report_v2.md</code> (full technical detail, methodology, and evidence trail)

This is a condensed version of `Final_Project_Report_v2.md`, written for a single reading in approximately 15–20 minutes. No technical finding has been removed — long narrative explanations have been replaced with tables, and repeated explanations have been cut. All figures are sourced from this project’s PostgreSQL database, source code, and `job_api.xlsx` research workbook (API Catalog and Sheet1). Where a fact could not be verified from these sources, this report states **”Not publicly available”** or **”Not verified during this project”** rather than estimating.

Contents

1. [Project Overview](#)
2. [Project Status](#)
3. [Implemented APIs](#)
4. [Complete Provider Research \(268 Providers\)](#)
5. [Internship Requirement Coverage](#)
6. [Final Recommendations](#)

1. Project Overview

Purpose. Build a single, de-duplicated, queryable dataset of job postings, sourced from as many legitimately accessible providers as possible, normalized into one schema with country and remote/hybrid/on-site classification, and persisted in a real database rather than a flat file.

Scope. The project researched **268 job-related APIs, ATS platforms, and job boards** (government labour agencies, commercial aggregators, ATS platforms, and job boards), narrowed that list to an **86-provider shortlist** for close audit, and built **46 working integrations** against a shared `sources/` interface (`BaseJobSource`). Every researched provider — implemented or not — is documented with a specific, evidence-based reason (Section 4).

Final outcome.

- **46 providers implemented; 41 currently contributing live data** to PostgreSQL.
 - **184,506 unique job records** stored, deduplicated by URL/hash, with per-provider run history.
 - **Offline country and work-arrangement normalization** — no paid geocoding service used.
 - The remaining **222 researched providers** were not implemented for specific, documented reasons: no public API, enterprise/partner-only access, wrong data-flow direction (employer-posting tools, not read APIs), deprecated services, or a registration/verification step not yet completed.
 - The platform reflects **one complete backfill run**, not yet a recurring schedule — it is ready for operational hardening, not further architecture, before production use.
-

2. Project Status

Area	Status	Detail
Provider research (268 researched, 86 shortlisted)	Completed	Every provider carries a documented reason for its outcome (Section 4)
46 provider integrations (shared <code>BaseJobSource</code> interface)	Completed	One module per provider under <code>sources/</code>
PostgreSQL persistence (<code>jobs</code> , <code>provider_runs</code> tables)	Completed	Upsert-on- <code>dedup_key</code> , idempotent schema
Deduplication (URL/hash-based)	Completed	0 cross-provider duplicate URLs remaining
Offline country normalization	Completed	Multi-country/region text forced to <code>NULL</code> rather than guessed
Work-arrangement classification (remote/hybrid/on-site)	Completed	Derived from location text and provider-native flags
<code>jobs.json</code> static export + frontend search/filter	Completed	Unchanged since Milestone 2
Per-source fault isolation	Completed	One broken provider cannot halt a run
41 of 46 providers contributing live data	Partially Completed	5 non-contributing for external reasons (billing, missing credentials, one zero-result run) — Section 3
Employment-type capture	Partially Completed	Present for 24 of 46 providers; absent for the rest
Country resolution long tail	Partially Completed	Multi-country case solved; ~1,565 distinct unresolved single-place values remain
Credentialed-provider setup	Partially Completed	2 of 12 credentialed providers lack credentials in this environment — Trade Me pending its approval process; CareerOneStop's registration is

Area	Status	Detail
		geographically restricted (geo-blocked) from this environment
6 real, accessible candidate providers (Saramin, VDAB, CBOP, Work24, GOV.UK Find a Job, Job-Room Switzerland)	Partially Completed	Registration/verification step outstanding, not rejected
Standardized job-category taxonomy	Not Started	No cross-provider taxonomy exists in the data (Section 5, item 11)
Salary / visa-sponsorship capture	Not Started	No schema field; no provider exposes it in a usable form
Scheduled/automated recurring syncs	Not Started	Every run to date has been a single manual invocation
Inline normalization at insert time	Not Started	Currently a separate, manually-triggered migration script
Provider-run health alerting	Not Started	No automated flag for a source dropping to 0 jobs
Dashboard/analytics UI	Not Started	Static search frontend only
Expired-job / posting-removal detection	Not Started	Requires a second ingestion run to diff against; none has occurred

3. Implemented APIs

3.1 LIFECYCLE SUMMARY

Stage	Count
Providers researched (API Catalog sheet, <code>job_api.xlsx</code>)	268
Providers shortlisted for close audit (Sheet1, <code>job_api.xlsx</code>)	86
Providers implemented (<code>sources/*.py</code> , registered in <code>sources/__init__.py</code>)	46
Providers currently contributing live data	41
Providers implemented but not currently contributing	5
Providers researched but not implemented	222

3.2 WHY 5 IMPLEMENTED PROVIDERS ARE NOT CURRENTLY LIVE

Provider	Reason
TheirStack	Account's billing plan does not include Jobs API access (HTTP 402)
Trade Me Jobs	Built, but no production credentials configured; also requires Trade Me's approval process
CareerOneStop	Built, but access/registration was geographically restricted (geo-blocked) from this environment, preventing successful API registration and testing — so <code>CAREERONESTOP_USER_ID</code> / <code>API_TOKEN</code> could not be obtained
Fantastic.jobs	Credentialed, but returned 0 jobs in the latest run — most plausibly a trial-window/account-tier issue, not confirmed
Arbeitnow	Free, no-auth API that previously worked (931 jobs in earlier testing); returned 0 jobs in the latest run — cause not confirmed

3.3 VOLUME CONCENTRATION

Provider	Jobs Stored	% of Total
SmartRecruiters	65,392	35.4%
Greenhouse	22,715	12.3%
Lever	17,980	9.7%
NHS Jobs	10,843	5.9%
USAJOBS	10,000	5.4%
Bundesagentur für Arbeit	9,896	5.4%
Reed	8,994	4.9%
NAV Arbeidsplassen	5,394	2.9%
Ashby	5,252	2.8%
Workable	4,137	2.2%

Top 3 providers supply 57.4% of all 184,506 stored jobs — volume is concentrated in ATS boards, not evenly spread across all 46 sources.

4. Complete Provider Research (268 Providers)

Tables 4.1 and 4.2 below satisfy the internship requirement to document the number of jobs available or retrieved for each API, alongside every provider's pricing, reference, and implementation status.

Every provider researched in this project (`job_api.xlsx` API Catalog sheet, 268 rows), split into **Implemented (46)** and **Not Implemented (222)**. Pricing model and payment terms are this project's own verified classification of each provider's published terms — where nothing could be confirmed, the cell reads "Not publicly available." Reference is a documented URL/domain found in this project's research notes.

4.1 IMPLEMENTED PROVIDERS (46)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Jobs Retrieved (Current Impl.)	Status	Implementation Notes
SmartRecruiters	No-auth public API	No	Free Public API	65,392 stored (65,394 fetched)	Live	• Limitation: 100 curated companies only, not a global index. • Expansion: Add more verified company slugs.
Greenhouse	No-auth public API	No	Free Public API	22,715 stored (22,715 fetched)	Live	• Limitation: 276 curated companies; single-shot fetch per company. • Expansion: Add more verified company slugs.
Lever	No-auth public API	No	Free Public API	17,980 stored (17,980 fetched)	Live	• Limitation: 162 curated companies; single-shot fetch per company. • Expansion: Add more verified company slugs.
NHS Jobs	No-auth public feed	No	Official Documentation (No Pricing Page)	10,843 stored (13,011 fetched)	Live	• Limitation: Undocumented endpoint; could change without notice. • Expansion: Monitor for endpoint changes; no code change needed.
USAJOBS	Free registration	No	Free Public API	10,000 stored (10,000 fetched)	Live	• Limitation: Hard 10,000-result API ceiling; no retry/error handling in fetch loop. • Expansion: Partition by JobCategoryCode/LocationName; add shared retry helper.
Bundesagentur für Arbeit	No-auth-style public key	No	Free Public API (undocumented; no official arbeitsagentur.de API docs exist)	9,896 stored (10,000 fetched)	Live	• Limitation: Hard 10,000-result API ceiling (page x size). • Expansion: Partition query by location/keyword params.
Reed	Free registration	No	Free Public API	8,994 stored (9,000 fetched)	Live	• Limitation: ~9,900 result-window boundary (HTTP 500 beyond); no shared retry helper. • Expansion: Raise page cap toward boundary; add shared retry helper.
NAV Arbeitsplätzen	No-auth-style public token	No	Free Public API	5,394 stored (12,478 fetched)	Live	• Limitation: Change-feed, not a current-postings search; drives high dup rate. • Expansion: Increase lookback window or page cap.
Ashby	No-auth public API	No	Free Public API	5,252 stored (5,252 fetched)	Live	• Limitation: 102 curated companies only. • Expansion: Add slugs; enable <code>includeCompensation=true</code> for salary data.
Workable	No-auth public API	No	Official Documentation (No Pricing Page)	4,137 stored (7,627 fetched)	Live	• Limitation: 89 curated companies; public widget API only, not full ATS. • Expansion: Add more verified company slugs.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Jobs Retrieved (Current Impl.)	Status	Implementation Notes
Workday	No-auth public API	No	Free Public API (no fixed docs URL; keyless, per-tenant endpoint)	2,997 stored (2,998 fetched)	Live	• Limitation: Self-capped at 200 jobs/tenant; large tenants under-sampled. • Expansion: Raise per-tenant page cap; add more tenants.
Himalayas	No-auth public API	No	Free Public API	2,253 stored (3,000 fetched)	Live	• Limitation: Self-capped at 3,000 of 90,000+; fixed page size (limit ignored). • Expansion: Raise page cap; use richer <code>/jobs/api/search</code> endpoint.
Arbetsförmedlingen	No-auth public API	No	Free Public API	2,096 stored (2,100 fetched)	Live	• Limitation: Hard offset $\leq 2,000$ ceiling (~2,100 of 40,000 reachable). • Expansion: None found; add in-loop error handling for resilience.
Adzuna	Free tier + paid metered	Yes, beyond free tier/quota	Official Documentation (No Pricing Page)	2,000 stored (2,000 fetched)	Live	• Limitation: Self-capped at 2,000/run; single-country (US) only. • Expansion: Raise page cap; query additional Adzuna country indexes.
The Muse	No-auth public tier	No	Free Public API	2,000 stored (2,000 fetched)	Live	• Limitation: Hard 100-page ceiling (HTTP 400) without an API key. • Expansion: Add an API key to raise the ceiling.
CareerJet	Free registration	No	Free Public API	1,996 stored (2,000 fetched)	Live	• Limitation: Self-capped at 2,000/run (no ceiling found in testing). • Expansion: Raise page cap.
EURES	No-auth public feed	No	Free Public API (undocumented)	1,681 stored (2,000 fetched)	Live	• Limitation: Hard ~10,000-result ceiling; self-capped at 2,000/run. • Expansion: Raise page cap toward the ~10,000 ceiling.
RemoteJobs.org	No-auth public API	No	Free Public API	1,500 stored (1,500 fetched)	Live	• Limitation: Real rate limiting observed (HTTP 429 ~page 23). • Expansion: None beyond current handling; partial results already preserved.
France Travail	Free self-service	No	Free Public API	1,146 stored (1,150 fetched)	Live	• Limitation: Self-capped range ceiling (1,150), inherited from prior research. • Expansion: Re-confirm true ceiling; raise if larger window confirmed.
Jooble	Free registration	No	Free Public API	1,137 stored (1,137 fetched)	Live	• Limitation: Usable window ~1,100-1,200 jobs vs. much larger reported totals; no in-loop retry. • Expansion: Add in-loop error handling.
Get on Board	No-auth public tier	No	Free Public API	1,011 stored (1,261 fetched)	Live	• Limitation: None significant; paid Private API tier unused by design. • Expansion: None identified; already fetches to each filter's total_pages.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Jobs Retrieved (Current Impl.)	Status	Implementation Notes
Taiwan Ministry of Labor	No-auth open data	No	Free Public API (data.gov.tw #44062)	1,000 stored (1,000 fetched)	Live	• Limitation: Hard 1,000-record cap; no working offset/page param found. • Expansion: None found; no further page accessible.
Findwork.de	Free registration	No	Free Public API	1,000 stored (1,000 fetched)	Live	• Limitation: Real rate limiting observed (HTTP 429 ~page 11). • Expansion: None beyond current handling; partial results already preserved.
Mustakbil.com	No-auth public feed	No	Free Public API	500 stored (500 fetched)	Live	• Limitation: No pagination exists; fixed 500-item feed. • Expansion: None found.
Teamtailor	No-auth public feed	No	Free Public API	494 stored (494 fetched)	Live	• Limitation: 25 curated companies; public RSS only, not credentialed Web API. • Expansion: Add more verified subdomains.
Recruitee (Tellent)	No-auth public API	No	Free Public API	394 stored (394 fetched)	Live	• Limitation: 21 curated companies; no pagination metadata in payload. • Expansion: Add more verified subdomains.
RemoteOK	No-auth public API	No	Free Public API	100 stored (100 fetched)	Live	• Limitation: No pagination exists; fixed ~100-job snapshot. • Expansion: None found.
MyJobMag	No-auth public feed	No	Free Public API	100 stored (100 fetched)	Live	• Limitation: No pagination (robots.txt blocks query strings); fixed 100 items. • Expansion: Add other MyJobMag country feeds (Ghana/Kenya/SA/UK).
Jobicy	No-auth public API	No	Free Public API	100 stored (100 fetched)	Live	• Limitation: No pagination; hard 100-job cap on count. • Expansion: None found.
We Work Remotely	No-auth public feed	No	Free Public API	99 stored (100 fetched)	Live	• Limitation: No pagination; fixed ~100-item feed. • Expansion: Add per-category feeds for broader coverage.
4dayweek.io	No-auth public feed	No	Free Public API	50 stored (50 fetched)	Live	• Limitation: No pagination; richer <code>/api/jobs</code> endpoint blocked by robots.txt. • Expansion: None; richer endpoint deliberately excluded.
SerpApi (Google Jobs)	Free tier + paid metered	Yes, beyond free tier/quota	Pricing Page	50 stored (50 fetched)	Live	• Limitation: Metered API; self-capped at 50 jobs/run to conserve quota. • Expansion: Raise page cap if a larger quota is approved.
OpenWebNinja (JSearch)	Free tier + paid metered	Yes, beyond free tier/quota	Pricing Page	49 stored (49 fetched)	Live	• Limitation: Metered API; self-capped at 5 pages to conserve quota. • Expansion: Raise page cap if a larger quota is approved.
Working Nomads	No-auth public API	No	Free Public API	33 stored (33 fetched)	Live	• Limitation: No pagination; fixed ~36-job snapshot; no RSS alternative found. • Expansion: None found.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Jobs Retrieved (Current Impl.)	Status	Implementation Notes
Remotive	No-auth public API	No	Free Public API	30 stored (30 fetched)	Live	• Limitation: No pagination; filter params confirmed non-functional. • Expansion: None found.
Freshersworld	No-auth public feed	No	Free Public API	30 stored (30 fetched)	Live	• Limitation: No pagination; rolling ~30-item window. • Expansion: None found.
Jobspresso	No-auth public feed	No	Free Public API	20 stored (20 fetched)	Live	• Limitation: Deeper pagination exists but blocked by robots.txt. • Expansion: None; deeper pagination deliberately unused.
MyCareersFuture	No-auth public API	No	Free Public API (undocumented)	20 stored (2,000 fetched)	Needs review	• Limitation: <code>limit</code> param ignored (fixed 20/page); self-capped at 2,000/run. • Expansion: Raise page cap; no hard API ceiling found.
NoDesk	No-auth public feed	No	Free Public API	10 stored (10 fetched)	Live	• Limitation: No pagination; fixed 10-item feed. • Expansion: None found.
Hasjob	No-auth public feed	No	Free Public API	6 stored (6 fetched)	Live	• Limitation: No pagination; inherently small volume (community board). • Expansion: None found.
HK Gov Vacancies	No-auth open data	No	Free Public API (data.gov.hk)	1 stored (58 fetched)	Needs review	• Limitation: Static dataset; no per-posting URL (collapses on dedup). • Expansion: Use a non-URL dedup key to reduce duplicate collapse.
TheirStack	Per-credit metered (1 credit/job)	Yes	Pricing Page	0 stored (0 fetched)	Blocked — billing	• Limitation: Bills per job returned; blocked by account billing (HTTP 402). • Expansion: Resolve billing; then raise job cap and add retry helper.
Trade Me Jobs	Free (registered app)	No	Official Documentation (No Pricing Page)	0 stored (0 fetched)	Built, uncredentialed	• Limitation: Built from API spec only, unexercised live; missing credentials (0 jobs). • Expansion: Provision consumer key/secret; complete production approval.
CareerOne Stop	Free registration	No	Free Public API	0 stored (0 fetched)	Built, uncredentialed	• Limitation: Hard ~750-result ceiling; access/registration was geographically restricted (geo-blocked) from this environment, preventing successful API registration and testing (0 jobs). • Expansion: Retry registration from an unrestricted network/region to obtain a user ID/API token.
Fantastic.jobs	Free trial → paid metered (~\$1/1,000 jobs)	Yes, after trial ends	Pricing Page	0 stored (0 fetched)	Credentialed, 0 result	• Limitation: Credentialed but 0 jobs; cause not confirmed. • Expansion: Verify account/trial status; add <code>/v1/active-jb</code> sibling endpoint.
Arbeitsnow	No-auth public API	No	Free Public API	0 stored (0 fetched)	0 in latest run	• Limitation: 0 jobs in latest run (cause unconfirmed); no retry/error handling.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Jobs Retrieved (Current Impl.)	Status	Implementation Notes
						• Expansion: Add retry/error handling; investigate 0-job run.

4.2 NOT IMPLEMENTED PROVIDERS (222)

Grouped by research classification at the time of this project. **Registration/Approval Required** and **Needs Live Verification** providers have a real, plausible access path (Section 6, Recommendations); **Not Viable** providers were rejected on structural grounds (no public API, enterprise-only, wrong data-flow direction, deprecated, or a dataset rather than a live API).

Registration/Approval Required (4)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
CBOP - Centralna Baza Ofert Pracy (Poland public employment offices)	Free	No	Free Public API	Potentially large - national public employment office coverage	Poland's national public-employment-office job database. Real SOAP web service returning batches of up to 1,000 live offers per file, but capped at 20 calls/cycle and only reachable during a 17:00-07:00 window -...
Saramin Open API (사람인 오픈 API)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Enterprise Contact / Pricing	Unknown - Saramin is one of Korea's largest commercial job boards but no public...	One of Korea's largest commercial job boards. Real documented API, but gated behind an application/approval step, explicitly labeled beta with a 500-calls/day cap - usable, just slower to unlock and rate-constrained.
VDAB Vacature API (Belgium - Flanders public employment service)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large - Flanders regional coverage	Flanders' public employment service vacancy API - free but requires an application form and an exploratory onboarding meeting with VDAB before credentials are issued. Real read API, just a slower registration path than...
Work24 (formerly Worknet/워크넷) Job Listings Open API - Korea Employment Information Service	Free	No	Not publicly available	Potentially large - South Korea's national public employment portal (renamed 고용24 in...	South Korea's national public job bank, verified live on data.go.kr with real-time XML/JSON updates and a free key. Caveat: license text restricts use to non-commercial, attributed, unmodified use - confirm terms with...

Needs Live Verification (4)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
GOV.UK Find a Job (DWP)	Free	No	Not publicly available	Potentially large	UK's national government job service; no confirmed public read/search API for third parties was found - only an employer-side SFTP bulk-upload path for posting vacancies. Worth one more direct check with DWP given its...
Job-Room Jobs API (SECO, Switzerland)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large	Switzerland's biggest job platform, but the API is built primarily for employers/ATS submitting ads under a reporting-obligation rule - whether third-party aggregators can get read-only access is unconfirmed in the...
JobsPikr	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Unknown (volume not published)	Commercial aggregator (incl. Naukri/Monster-sourced India feeds) with a real API and docs, but pricing is custom/quote-based rather than a published self-serve rate - confirm actual signup process and cost before...
Portal del Empleo / Servicio Nacional de Empleo (STPS Mexico)	Free	No	Not publicly available	Potentially large	Mexico's national employment service publishes datasets on the CKAN-based datos.gob.mx open-data platform, but a live, current-vacancy-specific API endpoint for the job board itself was not directly confirmed - may...

Not Viable at Research Time (214)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
104 Job Bank (104人力銀行)	Not applicable, no public API	Not applicable, no public API	104.com.tw	~1.1 million (110萬, Feb 2026)	No public API — site requires jobseeker/employer login; no developer access documented.
1111 Job Bank (1111人力銀行)	Not applicable, no public API	Not applicable, no public API	1111.com.tw	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
51job (前程无忧 / Qiancheng Wuyou)	Not applicable, no public API	Not applicable, no public API	51job.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
AMS eJob-Room (Austria public employment service)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
APEC (Association Pour l'Emploi des Cadres, France)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
APS Jobs (Australian Public Service careers)	Not applicable, no public API	Not applicable, no public API	apsjobs.gov.au	Moderate	No public API — site requires jobseeker/employer login; no developer access documented.
Adecco India	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Agencia Publica de Empleo SENA (APE)	Not applicable, no public API	Not applicable, no public API	edu.co	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
AllJobs.co.il	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	AllJobs.co.il	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Apideck	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Paid middleware re-wrapping ATSs (Greenhouse, Lever, Workday) already integrated directly for free.
Apify	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Apna	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large, blue-collar (board size; no public access)	No public API for third-party aggregators.
Appcast	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	appcast.io/how-programmatic-recruitment-advertising-improves-cost-per-hire/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Aura (by Aura Intelligence)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Proprietary workforce database spanning 20M+ companies (billions of data points;...	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
BMET / Ami Probashi (Bangladesh Overseas Employment)	Not applicable, no public API	Not applicable, no public API	amiprobashi.com	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
BOSS Zhipin (Kanzhun Limited, NASDAQ: BZ)	Not applicable, no public API	Not applicable, no public API	zhipin.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
BambooHR	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.
Bayt.com	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Bayt.com	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Bdjobs.com	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Bolsa Nacional de Empleo (BNE / SENCE, Chile)	Free	No	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Bossjob	Free	No	bossjob.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					documented.
Bright Data	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Free Public API	116.8M+ records (general Jobs dataset); LinkedIn Jobs dataset specifically 62.4M+	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.
BrighterMonday	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	brightermond ay.co.ke	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Broadbean (Veritone Hire)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	broadbean.c om/product- suite/job- distribution- platform/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Built In	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech/startup; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
Bumeran	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
CEIPAL	Not publicly available	Not publicly available	ceipal.com/pr icing	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
CV-Library	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
CWJobs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	CWJobs.co.u k	12,000+ jobs (per cwjobs.co.uk About Us page)	No public API — site requires jobseeker/employer login; no developer access documented.
CareerBuilder	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	careerbuilder. com/	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
CareerBuilder / Monster (Talent Network distribution)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	careerbuilder.com/talent-network	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
CareerLink Vietnam	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
CareerOne (Australia)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	careerone.com.au	100,000+ (live jobs)	Feed-ingest/distribution network only — no public read endpoint for consumers.
Careers24	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	careers24.com	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Careers@Gov	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (SG public sector; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
Catho Developer API	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
China MOHRSS National Public Employment Service Platform (全国公共招聘服务平台)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Civil Service Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
Claro Analytics (WilsonHCG)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	Real-time job-postings and 500M+ global talent profiles	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
Cliclavoro / Ministero del Lavoro Open Data (Italy)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Computrabajo	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Coresignal	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	399M+ multi-source records (incl. India)	Enterprise-tier data vendor sold via sales contact; no published self-serve price.
Cutshort	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (tech; board size not public)	No public API — site requires jobseeker/employer login; no developer access documented.
Darwinbox	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.
Datapeople (by Payscale)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	Not publicly available (N/A (analyzes customer job postings, not a broad aggregation ...	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
Dice	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech niche; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
Diffbot	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.
Draup	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	Global labor-market and job-postings data (aggregated multi-source)	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
Drushim	Not applicable to data access (fee is for employers posting jobs)	Not applicable to data access (see Pricing)	Drushim.co.il	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
	on this platform, not...	Model column)			
Egypt Ministry of Labour (formerly Ministry of Manpower) Job Portal	Not applicable, no public API	Not applicable, no public API	labour.gov.eg	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Elempleo	Not applicable, no public API	Not applicable, no public API	Not publicly available	Very large (CO; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
Eluta	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Employment Services of South Africa (ESSA) - Dept. of Employment and Labour	Free	No	labour.gov.za	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Forasna	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	forasna.com/employer/pricing	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Foundit (Monster India)	Not applicable, no public API	Not applicable, no public API	Not publicly available	~800k+ (board size; no public access)	No public API for third-party aggregators.
Freshteam	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.
Glassdoor Jobs API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	glassdoor.com/developer/index.htm	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Glints	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Google Cloud Talent Solution (Job Search API)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available (N/A (you populate it with your own jobs; not an aggregation ...	Hosted ML search engine for your own postings — not a source of third-party listings.
Google for Jobs	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	schema.org	Unknown (indexes open web)	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Greenwich.HR (WageScape)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	~5M new US jobs/month; ~70% of all open US jobs; 1.5M+ hiring organizations	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
GreytHR	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - HRMS/payroll only, no job feed)	No public API — site requires jobseeker/employer login; no developer access documented.
GulfTalent	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	gulftalent.com	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Gupy Public API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	gupy.io	~90,000 job vacancies published monthly on the Gupy R&S platform (official figure)	No public API — site requires jobseeker/employer login; no developer access documented.
HackerEarth Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (board size not public)	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
HasData	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Free Public API	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.
Hello Work Job Information API (ハローワーク求人情報提供サービス, MHLW)	Free	No	Not publicly available	Potentially large - nationwide public employment service covering hundreds of thousands ...	Real, documented API, but registration is restricted to licensed placement businesses/governments — a generic aggregator doesn't qualify.
HelloWork ATS Partner API (France)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Hired.com (LHH Recruitment Solutions)	Not applicable, no public API	Not applicable, no public API	Hired.com	0	No public API — site requires jobseeker/employer login; no developer access documented.
Hirist	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech; board size not public)	No public API — site requires jobseeker/employer login; no developer access documented.
Hong Kong Labour Department Interactive Employment Service (iES, jobs.gov.hk)	Not applicable, no public API	Not applicable, no public API	www.jobs.gov.hk	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Horsefly Analytics	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	1 trillion+ data points; millions of job postings; supply/demand and across 60+ countries	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
IEFPOne / netEmprego (Portugal public employment service)	Not applicable, no public API	Not applicable, no public API	Not publicly available	5,913 ofertas de emprego (live count)	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Indeed Hiring Lab	Free	No	Free Public API	Not publicly available (N/A (publishes indices/aggregates, e.g. Job Postings Index, Wage...	Indeed's research arm — publishes aggregate labor-market indices, not individual job records.
Indeed India	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Huge (not accessible - partner-only)	Enterprise partner-only access; not available to a generic aggregator.
Indeed PLUS Job Distribution API (Recruit Holdings, Japan)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Not publicly available	Potentially large - aggregates postings across Japan's major job boards distributed via...	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
InfoJobs (Spain)	Not applicable, no public API	Not applicable, no public API	Not publicly available	68,383 ofertas (live count, Spain, as of July 2026)	No public API — site requires jobseeker/employer login; no developer access documented.
Instahyre	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech; board size not public)	No public API — site requires jobseeker/employer login; no developer access documented.
Internshala	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large (intern/fresher; board size not public)	No public API — site requires jobseeker/employer login; no developer access documented.
Israeli Employment Service (Sherut HaTaasuka / taasuka.gov.il)	Free	No	taasuka.gov.il	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
JSearch (RapidAPI, by OpenWeb Ninja)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Free Public API	Not publicly available	Generic B2B data vendor, not a dedicated job board — paid, overlaps with sources already integrated.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Jadarat (Saudi National Employment Portal / HRDF)	Free	No	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Job Bank (ESDC) feed / Open Gov dataset	Free	No	Not publicly available	Very large (100,000+)	Live feed needs ESDC partner approval; fallback open dataset lacks employer names and apply URLs.
Job Market Finland / Tyomarkkinatori (TE-palvelut, Finland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
JobKorea	Not applicable, no public API	Not applicable, no public API	Not publicly available	~209,637 (live listings)	No public API — site requires jobseeker/employer login; no developer access documented.
JobStreet / JobsDB (SEEK group)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Very large if accessible (not public)	SEEK-group flagship boards for SEA — enterprise partner-only, no self-serve path.
JobThai	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	jobthai.com	~121,000 active jobs (121,337 shown live on jobthai.com)	No public API — site requires jobseeker/employer login; no developer access documented.
Jobberman	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	jobberman.com	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Jobcase (MaxRecruit)	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	jobcase.com/products/maxrecruit	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Jobcase Platform API	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	jobcase.com/	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Jobg8	Paid (metered/flat; see Cost Details column)	Yes, paid from the outset, no free tier	jobg8.com/Faq.aspx?contentid=10874	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
	source for exact terms)	documented			
Jobillico	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Jobindex (Denmark)	Not applicable, no public API	Not applicable, no public API	Not publicly available	33,600 jobs (live) / 30,000+ job ads	No public API — site requires jobseeker/employer login; no developer access documented.
JobisJob (LIFULL Connect)	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	jobboardfinder.com/jobboard-jobisjob-usa	~9,000,000 (monthly, cross-country)	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Jobnet.dk (Denmark public employment portal)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Jobrapido	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	jobrapido.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Jobs.cz	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (CZ/SK; not quantified)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.
JobsDB Hong Kong (SEEK Group) / CTgoodjobs	Not applicable, no public API	Not applicable, no public API	hk.jobs	~32,700 (JobsDB HK) / ~32,000 (CTgoodjobs)	No public API — site requires jobseeker/employer login; no developer access documented.
JobsEQ (Chmura Economics & Analytics)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	Real-time job-posting dataset with international coverage	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
JobsIreland.ie (Ireland public)	Not applicable, no public API	Not applicable,	Not publicly available	4,806 vacancies (live count)	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
employment service)		no public API			
Jobsite	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Jobsite.co.uk	280,000+ live job adverts (per jobsite.co.uk About Us page)	No public API — site requires jobseeker/employer login; no developer access documented.
JoinVision (JobCloud HR Tech)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	crunchbase.com/organization/joinvision	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Jora (Australia)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	jora.com	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Joveo	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	joveo.com/pr-ogrammatic-job-advertising-platform/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
JustRemote	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	justremote.co/remot-jobs	2,000+ hidden remote jobs	No public API — site requires jobseeker/employer login; no developer access documented.
Kalibr	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Karir.com	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Keka	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - HRMS/payroll only, no job feed)	No public API — site requires jobseeker/employer login; no developer access documented.
Kelly Services India	Not applicable, no public API	Not applicable,	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
		no public API			documented.
Kemnaker Karirhub (SIAPkerja)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Kombo.dev	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Pricing Page	Not publicly available	Paid middleware re-wrapping ATSS (Greenhouse, Lever, Workday) already integrated directly for free.
Ladders	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (niche; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
Lensa	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	lensa.com/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Lightcast (formerly Emsi Burning Glass) Job Postings API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	Billions of postings aggregated from 160,000+ sources (18B+ labor-market data points)	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
Lightcast Jobs Canada	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Not publicly available	Paid labor-market-intelligence product sold via sales process, not self-serve.
LinkUp	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	Millions of postings indexed daily from 80,000+ employer websites	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
LinkedIn Jobs	Enterprise	Yes, enterprise/s	Not publicly available	Huge (not accessible -	Enterprise partner program only — no self-serve developer signup.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
		ales pricing, no self-serve free tier		partner-only)	
MOHRE UAE Federal Careers / iRecruitment Portal	Not applicable, no public API	Not applicable, no public API	mohre.gov.ae	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
MP Rojgar	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
MahaJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
ManpowerGroup India	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Mantiks	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic B2B data vendor, not a dedicated job board — paid, overlaps with sources already integrated.
Merge.dev	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Paid middleware re-wrapping ATSs (Greenhouse, Lever, Workday) already integrated directly for free.
Mitula (LIFULL Connect / Adzuna)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	lifullconnect.com/brands/mitula/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Monster	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	monster.com/solutions/talent-management/partner-with-monster/	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
NAVTTTC National Employment Exchange Tool (NEXT / jobs.gov.pk)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
NEAIMS - National Employment Authority Kenya	Free	No	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
National Career Service (NCS)	Free	No	Free Public API	Large in aggregate	Periodic bulk open-government dataset, not a live per-job search API — stale snapshots only.
National Directorate of Employment (NDE) Nigeria	Not applicable, no public API	Not applicable, no public API	nde.gov.ng	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
National Job Portal (NJP) - Pakistan	Not applicable, no public API	Not applicable, no public API	njp.gov.pk	Small-to-moderate	No public API — site requires jobseeker/employer login; no developer access documented.
National Job Portal - Bangladesh (jobs.gov.bd)	Not applicable, no public API	Not applicable, no public API	jobs.gov.bd	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Naukri	Not applicable, no public API	Not applicable, no public API	Not publicly available	~1M+ (board size; no public access)	No public API; any access is undocumented/grey-area and India-IP-restricted.
Naukrigulf	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	naukrigulf.com	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Nexxt (formerly Beyond.com)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Beyond.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Nimble (Nimbleway)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.
OCC Mundial	Not applicable, no public API	Not applicable, no public API	Not publicly available	137,925 active vacancies ("vacantes activas en occ.com.mx", as of Oct 18, 2024);...	No public API — site requires jobseeker/employer login; no developer access documented.
Outsourcely	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	outsourcely.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Oxylabs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
PandoLogic (pandoIQ) / Veritone Hire	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	pandologic.com/solutions/programmatics-advertising-pandoiq/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Pangian	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	pangian.com/feed/?post_type=job_listing	Varies (curated remote listings)	Site redirects to a static shutdown notice — service confirmed discontinued, not merely credential-gated.
People Data Labs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Free Public API	Not publicly available	Generic B2B data vendor, not a dedicated job board — paid, overlaps with sources already integrated.
PhilJobNet (DOLE Bureau of Local Employment)	Free	No	philjobnet.gov.ph	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Piloterr	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.
Pnet	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	pnet.co.za	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Portal Empleo Argentina (Ministerio de Capital Humano / ex-Trabajo)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Pracuj.pl (Poland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	762,827 job offers published in 2025	No public API — site requires jobseeker/employer login; no developer access documented.
PredictLeads	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	270M+ records since 2018; ~9.8M active jobs	Generic B2B data vendor, not a dedicated job board — paid, overlaps with sources already integrated.
Profesia	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (CZ/SK; not quantified)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Public Service Commission of Sri Lanka (psc.gov.lk)	Not applicable, no public API	Not applicable, no public API	psc.gov.lk	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Qess Corp	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Radancy	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	radancy.com/en/programmatic-adtech/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Randstad India	Not applicable, no public API	Not applicable, no public API	Not publicly available	1,394 jobs (live count on official site)	No public API — site requires jobseeker/employer login; no developer access documented.
Recruit CRM	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.
Recruitics	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	recruitics.com/programmatic-job-advertising	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Remote First Jobs (Dynamite Jobs)	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	remotefirstjobs.com/rss	Varies (curated remote-first listings)	Blocked by an active Cloudflare JS challenge on every path — a technical barrier, not a credentials gate.
Remote.co	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	remote.co/feed	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Revelio Labs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	1.1B+ workforce profiles plus job postings, transitions,	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
				sentiment, layoffs	
Rikunabi / Mynavi (Recruit Co. / Mynavi Corporation)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Rojgar Sangam (UP)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
Rozee.pk	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
SEEK (SEEK.com.au / SEEK.co.nz)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	SEEK.com.au	Potentially large	Feed-ingest/distribution network only — no public read endpoint for consumers.
SEPE Open Data Portal (Spain public employment service)	Free	No	Not publicly available	Not publicly available	Publishes only static statistical open-data files — no live REST API for current vacancies.
SINE Aberto (Sistema Nacional de Emprego / Ministerio do Trabalho e Emprego)	Free	No	Free Public API	Potentially large	Suspended since Oct 2022 pending LGPD compliance review — no active API currently.
ScraperAPI	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.
Scrapingdog	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.
Shine	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (board size; no public access)	No public API — site requires jobseeker/employer login; no developer access documented.
SmartDreamers	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	smartdreamers.com/pricing	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Snagajob	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	snagajob.com/	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
StepStone	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	stepstone.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Superset	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (campus; per-institution, not aggregatable)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.
Talent.com	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Talent.com	40 million+ jobs (global figure stated on talent.com homepage)	No public API — site requires jobseeker/employer login; no developer access documented.
TalentNeuron (Gartner)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	Real-time labor data covering ~90% of world GDP; 40TB normalized data, 3B+ profiles,...	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
Talroo	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	talroo.com/pr/	Millions (per Talroo publisher marketing)	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
TeamLease	Not applicable, no public API	Not applicable, no public API	Not publicly available	23,082+ active job vacancies (live count on official site)	No public API — site requires jobseeker/employer login; no developer access documented.
Techmap.io	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Free Public API	292M+ available jobs	Generic B2B data vendor, not a dedicated job board — paid, overlaps with sources already integrated.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Telangana T-Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No public API — site requires jobseeker/employer login; no developer access documented.
Textkernel Jobfeed / Market IQ	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Enterprise Contact / Pricing	Large multi-country vacancy corpus (millions of postings, deduplicated/enriched)	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
Thailand Department of Employment - Smart Job Center (ไทยแรงงานพา)	Free	No	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
The Burning Glass Institute	Free	No	Free Public API	Not publicly available (N/A (research using Lightcast and other data))	Enterprise labor-market-intelligence SaaS sold via sales contract, not a self-serve jobs feed.
TimesJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (board size; no public access)	No public API — site requires jobseeker/employer login; no developer access documented.
TopCV	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
TotalJobs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Totaljobs.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Triplebyte	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	triplebyte.com/	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Trovit (LIFULL Connect / Adzuna)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	trovit.com/fee-d-in-jobs/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
UWV Open Match Data (Netherlands)	Free	No	Not publicly available	Not publicly available (N/A - aggregated statistics only, not individual postings)	Publishes aggregate labor-market statistics only — no individual job postings or per-job ID.
Unified.to	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Not publicly available	Not publicly available	Paid middleware re-wrapping ATSs (Greenhouse, Lever, Workday) already integrated directly for free.
Unstop	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (campus/intern; not a job-listing API)	No public API — site requires jobseeker/employer login; no developer access documented.
Upward.net (a Jobcase company)	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	upward.net/employers	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Vagas for Business API (Vagas.com.br)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Vagas.com	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
VietnamWorks Open API	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Enterprise Contact / Pricing	Potentially large	Partner access required — gated to approved commercial partners, not open registration.
WUZZUF	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	wuzzuf.net	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.
Welcome to the Jungle	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Wellfound	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (startup roles; board	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
				size not public)	
WhatJobs	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	whatjobs.com/job-aggregators	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
Work and Income / Find a Job (MSD, New Zealand)	Not applicable, no public API	Not applicable, no public API	workandincome.govt.nz/	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Work at a Startup (Y Combinator)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	workatastartup.com	Unknown (thousands across 1,000+ YC companies)	No public API — site requires jobseeker/employer login; no developer access documented.
WorkIndia	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large, blue-collar (board size; no public access)	No public API — site requires jobseeker/employer login; no developer access documented.
Workato	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Pricing Page	Not publicly available	Paid middleware re-wrapping ATSS (Greenhouse, Lever, Workday) already integrated directly for free.
Workforce Australia	Free	No	Free Public API	Potentially large	API is write/management-only (employers post via OAuth2) — no read/search endpoint exists.
Workopolis	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
WowJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Xing Jobs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
XpressJobs (xpress.jobs)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
ZipAlerts (ZipRecruiter TrafficBoost)	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	zipalerts.com/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
ZipRecruiter API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Large (not quantified)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.
Zoho Recruit	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-only access — gated behind sales/partner approval, no self-serve path.
ZonaJobs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	zonajobs.com.ar	Not publicly available	No public API — site requires jobseeker/employer login; no developer access documented.
Zyte (formerly Scrapinghub)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Pricing Page	Not publicly available	Generic web-scraping infrastructure, not a job-specific API — would require building a custom scraper.
data.gov.in OGD platform	Free	No	Not publicly available	Varies (per-resource)	General-purpose open-government-data portal (33 sectors), not jobs-specific — static datasets, not a live feed.
eQuest	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	equest.com/	Not publicly available	Advertiser-side distribution network — built for employers to push jobs out, not for aggregators to pull data in.
iimjobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (management roles; board size not public)	No public API — site requires jobseeker/employer login; no developer access documented.
jobs.ch (Switzerland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	45,825 jobs (live)	No public API — site requires jobseeker/employer login; no developer access documented.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
karriere.at (Austria)	Not applicable, no public API	Not applicable, no public API	github.com/karriere.at	13,400+ jobs (live)	No public API — site requires jobseeker/employer login; no developer access documented.
topjobs.lk	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public API — site requires jobseeker/employer login; no developer access documented.

5. Internship Requirement Coverage

2. FINAL PROJECT REPORT

This document, together with the companion technical report

`Final_Project_Report_v2.md`, is the final project report.

3. APIS/SERVICES USED

46 providers implemented (Section 3); 268 researched in total (Section 4).

4. COMPLETED VS. PENDING

See the [Project Status table](#) (Section 2).

5. FULLY / PARTIALLY / NOT STARTED

See the [Project Status table](#) (Section 2).

6. PRICING MODEL (FREE / FREE TRIAL / PAY-AS-YOU-GO / SUBSCRIPTION / ENTERPRISE)

Pricing Model (Implemented Providers)	Count	Providers
Free (no cost, ever)	41	40 fully free/no-auth or free-registration providers, plus Trade Me Jobs (free, but approval-gated)
Free Trial (time-limited, then paid)	1	Fantastic.jobs (7-day trial, then metered)
Pay-as-you-go / Metered (has a paid, usage-based component)	5	Adzuna, SerpApi, OpenWebNinja, TheirStack, Fantastic.jobs
Subscription (flat recurring fee)	0	Not used by any implemented provider
Enterprise (sales-negotiated contract)	0	Not used by any implemented provider — 15 enterprise-only candidates were identified in research and not implemented (Section 4.2)

Full per-provider pricing model is in Section 4.1. The 222 not-implemented providers' pricing (where verifiable) is in Section 4.2.

7. PAYMENT REQUIRED AFTER FREE TIER

Answer	Count (of 46 implemented)
No (free with no paid tier)	41
Yes, beyond a free tier/quota, or after a trial	5

8. PRICING / DOCUMENTATION LINKS

All 46 implemented providers have an official reference URL, quoted directly from source code (Section 4.1). Of the 222 not-implemented providers, a documented reference (domain or URL) was found in this project's research for **166 of 268**

providers overall; for the remainder, the reference cell reads “Not publicly available” rather than a guessed link (Section 4.2).

9. NUMBER OF JOBS

Metric	Value
Total unique jobs stored (PostgreSQL)	184,506
Total jobs fetched, all providers, pre-dedup	200,727
Distinct companies represented	18,367

Top providers by volume are in Section 3.3. Approximate job counts for not-implemented providers (where publicly known) are in Section 4.2.

10. REGION COVERAGE

Region	Dedicated Providers	Notes
United States	4	USAJOBS, CareerOneStop, Findwork.dev, Adzuna
Canada	0	No dedicated source; incidental coverage only via global ATS boards
United Kingdom	2	Reed, NHS Jobs
Europe (non-UK)	5	Bundesagentur für Arbeit, Arbetsförmedlingen, France Travail, EURES, NAV Arbeidsplassen
India	2	Freshersworld, Hasjob
LATAM	1	Get on Board
APAC	4	MyCareersFuture, Taiwan MOL, HK Gov Vacancies, Trade Me
Africa	1	MyJobMag
Middle East	0	No single-country dedicated source (Mustakbil spans 20+ countries, counted as global)
Global / Remote / Multi-country	27	All 6 ATS platforms plus 21 aggregator/remote-board providers

Resulting job distribution is US-heavy (47.3%), then UK (14.3%) and Germany (6.9%) — a sourcing-mix effect from which countries have dense curated ATS company lists, not a normalization defect. Canada remains the clearest unresolved regional gap.

11. JOB CATEGORIES

No standardized, cross-provider job category exists in this dataset, and this cannot be supported without inventing one.

Verified directly against all 46 providers' `normalize()` implementations:

- **0 of 46** providers expose a job-category field that means the same thing across providers.
- **23 of 46** expose a provider-specific department, team, business-unit, occupation, or function field (e.g. SmartRecruiters' `function`, Taiwan Ministry of Labor's occupation code, USAJOBS's federal job series) — each proprietary to that one provider.
- **23 of 46** expose no category-like field at all (confirmed in code, e.g. Reed's module comment: "no job-category/employment-type field at all").

Evidence that these fields cannot be merged: the stored tag "Engineering" comes from SmartRecruiters' `function` field (6,797 jobs), Greenhouse's `departments` field (1,154 jobs), and Ashby's `department` / `team` field (1,136 jobs) — three structurally unrelated fields that happen to use the same English word. Treating them as one taxonomy would silently merge unrelated data. This project's schema stores all of this in a single free-text `tags` array; a real crosswalk would require manually mapping each provider's raw values, which has not been done.

12. DATA QUALITY

Dimension	Finding
Missing fields	Confirmed per-provider in code — e.g. Reed exposes no employment-type field at all; no provider exposes usable salary data
Company info	43 of 46 providers give a direct company-name field; 3 derive it (Ashby, Lever, Workday)
Duplicate rate	8.08% overall (see item 14 below)
Stale postings	11.9% of stored jobs are over 1 year old

13. JOB FRESHNESS

Metric	Value
Jobs with a posted date recorded	184,129 of 184,506 (99.8%)
Median posting age	22 days
Mean posting age	229 days (skewed by a long tail of old postings on some boards)
Posted within 30 days of snapshot	108,003 (58.6%)
Older than 1 year	21,915 (11.9%)

Refresh frequency and expired-job detection are **not verified during this project** — the dataset reflects one ingestion run, not a recurring schedule, so there is no interval or diff to measure yet.

14. DUPLICATE ANALYSIS

Metric	Value
Total in-run duplicates skipped (all providers)	16,221
Total fetched, pre-dedup	200,727
Overall deduplication rate	8.08%
Cross-provider duplicate URLs remaining	0

Highest duplicate rates: MyCareersFuture (99.0%, `limit` param silently ignored by the provider), HK Gov Vacancies (98.3%, every job shares one generic portal URL), NAV Arbeitsplassen (56.8%, a change-feed rather than a current-postings search), Workable (45.8%, the same job posted to multiple cities, correctly collapsed).

How deduplication actually works. `dedupe_jobs()` (`jobs.py`) and `dedup_key_for()` (`database.py`), enforced via `ON CONFLICT (dedup_key)` collapse jobs whose **URL string matches exactly** (or, when a provider has no URL, a hash of its identifying

fields). This is why “cross-provider duplicate URLs remaining” is 0: that figure measures literal URL collisions, not whether the same real-world job posting appears under two different URLs from two different providers. The true overlap rate between architecturally distinct providers has not been measured in this project, and no percentage is estimated here — doing so would require fuzzy matching on (title, company, location) across providers, which has not been built.

Where duplicate/overlapping jobs are likely to occur, and why:

- **Single-tenant company ATS boards (Greenhouse, Lever, Ashby, Workday, SmartRecruiters, Teamtailor, Recruitee, Workable)** — each is queried per-company against a curated slug list (Section 4.1), so a job appears once, under that company’s own ATS-hosted URL. Overlap *among these providers* is structurally unlikely in this project, since no company slug is deliberately queried against two different ATS platforms at once. The risk instead comes from the next category: if the same employer’s postings are also picked up by a general aggregator, the aggregator’s copy carries a different URL (its own listing page or tracking redirect), so today’s exact-URL dedup would not catch the overlap. (Two other well-known ATS platforms not implemented here — BambooHR, researched in Section 4.2 as enterprise/sales-gated, and Jobvite, not part of this project’s 268-provider catalog — would carry the identical structural risk if added.)
- **Aggregators that re-publish employer-direct postings (Adzuna, Reed, Jooble)** — these are commercial aggregators that crawl or license listings from many original sources, which can include employers whose ATS boards (Greenhouse, Lever, Workday, etc.) this project already queries directly. A job posted by a Greenhouse-hosted employer could plausibly also surface through Adzuna, Reed, or Jooble’s feed under a different URL, invisible to exact-URL dedup.
- **Government/labour-exchange boards (USAJOBS, Bundesagentur für Arbeit, NAV Arbeidsplassen, EURES, France Travail)** — some of these mirror each other by design (EURES federates postings from national labour exchanges, including Bundesagentur and NAV Arbeidsplassen), so the same vacancy can legitimately appear on more than one government board, each with its own URL. USAJOBS is architecturally separate (US federal listings only) and not expected to overlap with the European boards.
- **Why deduplication is required.** Uncaught duplicates would inflate stored job counts, skew per-provider volume comparisons (Section 3.3), and double-count a company’s hiring activity in any downstream analytics.

- **How the current implementation handles it.** Exact-URL (or identifying-field-hash) matching at insert time, both within a single run (`dedupe_jobs`) and across runs/providers via `dedup_key` . This correctly and completely handles same-URL duplicates (confirmed: 0 remaining) and same-job-different-city listings within one provider (e.g., Workable’s 45.8% above). It does **not** catch the same job posted under different URLs by two different providers — that requires fuzzy/semantic matching on (title, company, location), which is not implemented and is listed as future work (Section 6).

Exact overlap percentages between specific provider pairs (e.g., what share of Adzuna’s UK jobs also appear via Reed) were not calculated in this project and are not estimated here, since the current dedup logic has no way to measure them.

15. RATE LIMITS

What these metrics mean. Providers describe their limits in different units; all of the following appear across this project’s research notes and implementation:

- **Requests per second (QPS)** — the maximum call rate at any instant; exceeding it triggers immediate throttling (typically HTTP 429).
- **Requests per minute / hour** — a rolling or fixed window cap, most relevant to pacing a single fetch loop against one provider.
- **Requests per day / month** — a longer-cycle quota, usually tied to a free tier or metered billing plan (e.g., SerpApi/OpenWebNinja’s 200 requests/month below).
- **Unlimited** — no documented ceiling for the endpoint used, though undocumented soft limits can still surface in practice (e.g., Reed’s ~9,900-record ceiling below, discovered by testing rather than published in any doc).
- **Burst limits** (where applicable) — a short-window allowance above the steady-state rate (e.g., “X req/s, burst to Y”) that lets a client briefly exceed the normal ceiling before throttling engages; not separately documented by any of the 46 implemented providers in this project.

Limit type	Examples
Hard result-window ceiling	USAJOBS and Bundesagentur für Arbeit (10,000 records); Reed (~9,900, returns HTTP 500 beyond it)
Monthly quota	SerpApi and OpenWebNinja (200 requests/month free tier)
Per-job billing	TheirStack (1 credit per job returned)

The examples above are the limits discovered and documented during implementation (`Final_Project_Report_v2.md`, Section 7); not all 46 providers' rate limits were independently re-verified for this report.

16. RESPONSE TIME & RELIABILITY

Provider	Duration (s)	Reliability
SmartRecruiters	385.7	High
Lever	870.1	High (slowest run despite modest volume — per-company pagination overhead)
Greenhouse	183.8	High
NHS Jobs	102.4	Undocumented endpoint (no official API contract)
USAJOBS	53.1	High
Bundesagentur für Arbeit	174.3	High (with retry logic)
Reed	56.6	High
NAV Arbeitsplassen	51.4	Rotating token (no fixed key)
Ashby	70.3	High
Workable	51.8	High

Full duration/reliability figures for all 46 providers exist in the underlying `provider_runs` table; only the ten highest-volume providers are shown here for brevity. The observations below are based on this project's implementation behaviour (source code plus the one completed backfill run), not formal or repeated benchmarking — no provider was load-tested or measured across multiple runs.

What drives duration. All 46 sources run **sequentially, not in parallel** —

`collect_all_jobs()` (`jobs.py`) loops over `SOURCES` one at a time, so today's total run time is the sum of every provider's individual duration, and any one slow or retrying provider delays the whole run rather than overlapping with the others. This is the single biggest lever for a production cron schedule: parallelizing independent providers (they share no state besides the final combined job list) would let total wall-clock time approach the slowest single provider instead of the sum of all 46.

- **Fast providers** — sources with a single request or a small, fixed number of pages return in well under a minute (e.g. NAV Arbeidsplassen, USAJOBS, Reed, and Workable above, all under ~60s).
- **Slow providers** — Lever is the slowest observed run (870.1s) despite only 17,980 jobs stored, because it paginates per-company across its curated company list; SmartRecruiters (385.7s) is slower in absolute terms but fetches the most jobs of any provider (65k+), so its per-job cost is actually low.
- **Providers requiring retries** — 36 of 46 providers call the shared `request_with_retry()` helper (`sources/utils.py`), which retries connection errors, timeouts, and HTTP 429/500/502/503/504 with exponential backoff (2s, 4s, 8s across up to 3 attempts); a few others (e.g. Bundesagentur für Arbeit) implement their own retry logic rather than the shared helper. A provider that trips this path runs measurably slower purely from backoff sleep time.
- **Providers affected by rate limits** — RemoteJobs.org and Findwork.dev both hit real, observed HTTP 429 responses during implementation (Section 4.1 notes), confirming the retry path is exercised in practice and not just defensive code. See Section 15 for the other documented ceilings (result-window caps, monthly quotas, per-job billing) that affect specific providers' fetch loops.

- **Providers requiring authentication** — 15 of 46 providers require some credential (API key, OAuth, or HTTP Basic; Section 17). These carry a failure mode the other 31 don't: if a key expires, rotates, or hits a quota, that provider's run can fail independently of network conditions, as already observed for CareerOneStop, Trade Me, and TheirStack (Section 3.2).
- **Sequential execution (current state)** — one provider at a time, in the fixed order of `sources/__init__.py`'s `SOURCES` list; no concurrency between providers today.
- **Parallel execution (recommended change)** — running independent providers concurrently (e.g., a worker pool keyed by provider) is the most direct way to shorten a production run; each provider's own internal pagination should stay sequential, since none of these APIs were built for concurrent pagination against the same key/session.
- **Expected impact on production scheduling** — because this project reflects one manual run, not a recurring schedule (Section 2), no production interval has been established yet. Given sequential execution and the durations observed (up to 870.1s for the slowest single provider), a naive sequential cron job across all 46 providers should be expected to take on the order of tens of minutes end-to-end; the exact figure depends on which providers are retrying or rate-limited at run time, which varies run to run. Parallelizing providers, and isolating credentialed/rate-limited providers from the no-auth majority (so one slow retry doesn't block unrelated fast providers), are the two changes most likely to shorten and stabilize a production schedule.

17. AUTHENTICATION

Auth Type	Count (of 46)
None (no auth required)	31
API key (registered)	10
Public non-secret key/token (no registration)	2
OAuth	2
HTTP Basic	1

18. FILTERING CAPABILITIES

Provider	Filtering Available
SmartRecruiters	Company, offset/limit
Greenhouse	Company only
Lever	Company, offset/limit
NHS Jobs	Keyword
USAJOBS	Page/ResultsPerPage only (no keyword/filter params implemented)
Bundesagentur für Arbeit	Keyword, location
Reed	Pagination only (no keyword/filter params implemented)
NAV Arbeidsplassen	Modified-since cursor
Ashby	Company only
Workable	Company only

Filtering capability was verified for the ten highest-volume providers; it was **not independently documented for the other 36 implemented providers** in this report. No provider was found to expose salary, experience-level, or visa-sponsorship filtering.

19. MUST HAVE / SHOULD HAVE / NICE TO HAVE

Priority	Item
Must Have	Move normalization inline into the save path
Must Have	Scheduled, automated recurring syncs
Must Have	Resolve the 5 non-contributing providers (credentials, billing, root-cause)
Should Have	Reporting dashboard on top of PostgreSQL
Should Have	Better location enrichment (~1,565 distinct unresolved country values)
Should Have	Normalized employment-type field; a department/occupation crosswalk table
Should Have	Dedicated Canada, broader APAC, and Middle East sources
Should Have	Provider-run health alerting
Should Have	Complete the 6 deferred candidate providers (Saramin, VDAB, CBOP, Work24, GOV.UK Find a Job, Job-Room Switzerland)
Nice to Have	Funding-stage/company-size enrichment
Nice to Have	Salary capture where a source exposes it
Nice to Have	Historical trend tracking across multiple runs
Nice to Have	Periodic re-verification of the 268-provider research catalog

20. PRODUCTION RECOMMENDATIONS

Summarized here; full detail in Section 6. Before production reliance: close the 5-provider contribution gap, add provider-run health alerting, move normalization inline, and put the pipeline on a recurring schedule.

6. Final Recommendations

Current readiness. The platform is functionally complete for a first release: 46 providers integrated, 184,506 deduplicated jobs stored in PostgreSQL, offline country/work-arrangement normalization applied, and full fault isolation so one broken provider cannot take down a run. It reflects **one complete backfill run**, not an ongoing sync.

Remaining work, in priority order:

1. Provision missing credentials (Trade Me), retry CareerOneStop's registration from a network/region that is not geo-blocked, resolve TheirStack's billing block, verify Fantastic.jobs' trial/account status, and re-run Arbeitnow to confirm its 0-job result was transient.
2. Pursue the two highest-value deferred candidate providers — Saramin and VDAB — both real APIs blocked only by a registration/onboarding step.
3. Add provider-run health alerting before scheduling recurring syncs, so a source silently dropping to 0 doesn't go unnoticed.
4. Move country/work-arrangement normalization inline into the ingestion path, replacing the current manual migration script.
5. Investigate the two low-yield sources (HK Gov Vacancies, MyCareersFuture) before relying on either for APAC coverage.

Production readiness. Not yet production-scheduled. Recommended before relying on this pipeline for ongoing reporting: (a) resolve the 5-provider gap above, (b) add health alerting, (c) put the pipeline on a recurring cron/schedule, (d) confirm secrets handling carries through to the deployment target, (e) extend automated test coverage from country normalization to the full aggregation/dedup pipeline.

Next steps. Treat the items above as a scoped, well-understood backlog rather than unknowns — none of them block the current dataset from being useful today. The full technical detail, evidence, and provider-by-provider reasoning behind every recommendation is in `Final_Project_Report_v2.md`.

Job Aggregation Platform — Final Project Report · Data snapshot 2026-07-10 · Report date July 15, 2026 · Condensed from `Final_Project_Report_v2.md` for a single 15–20 minute engineering-manager review. No technical finding was removed; long narrative sections were replaced with tables, and repeated explanations were cut. The Implemented Providers table (Section 4.1) was independently re-verified against `sources/*.py` on 2026-07-15, adding per-provider implementation limitations and expansion opportunities; all 46 providers' Official Reference links were additionally re-checked live against each provider's own domain on 2026-07-15, replacing any dead link, redirected domain, or non-official (e.g. third-party GitHub wrapper) reference with the correct official URL. Source of truth: PostgreSQL (`jobs` , `provider_runs`), this repository's source code, and `job_api.xlsx` (API Catalog, Sheet1 sheets). Where a fact could not be verified, this report states "Not publicly available" or "Not verified during this project" rather than estimating.